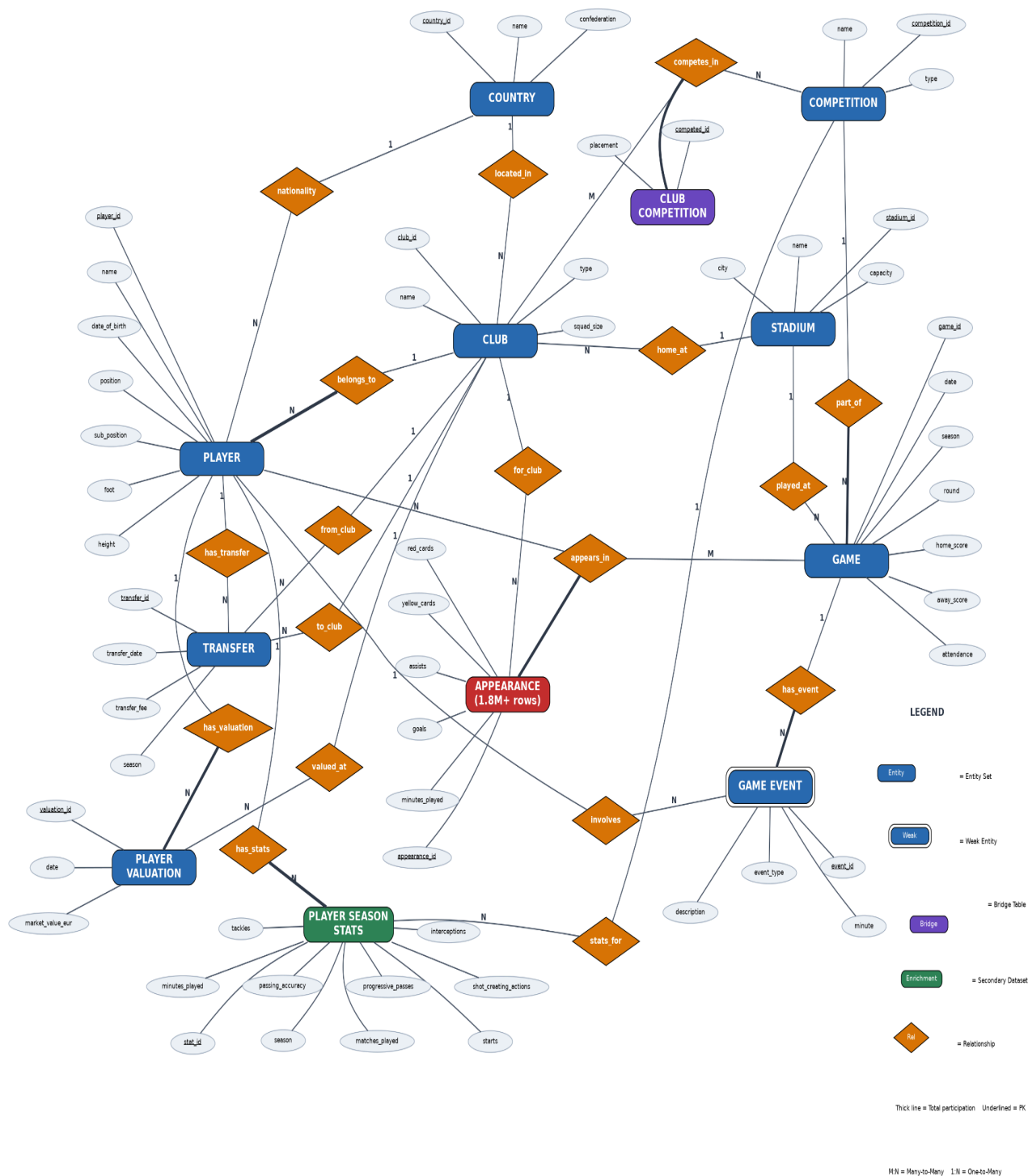


## Checkpoint 2

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### 1. Revised ER Diagram



The ER diagram has been revised since Checkpoint 1. Here are the changes we made based on instructor feedback and our own review:

Stadium pulled out into its own entity. We had stadium showing up in both Club and Game, which was flagged as redundant. Now there's a separate Stadium entity (with name, city, capacity), and both Club and Game just reference it through foreign keys.

GameEvent relationships made explicit. There was a question about how GameEvent differs from Competition. They're completely different things -- Competition is a league or tournament, GameEvent is something that happens during a match (a goal, a card, a sub). We added clear relationship lines from GameEvent to Game and Player so this is obvious in the diagram.

Added PlayerSeasonStats for the secondary dataset. The FBref data (passing accuracy, shot-creating actions, tackles, etc.) wasn't represented anywhere in CP1. We added a new entity for it, connected to both Player and Competition since stats are tracked per player per league per season.

ClubCompetition bridge table added. The Club-Competition many-to-many was just a line in CP1. Now it's an explicit bridge table with a placement attribute to track league finishes.

Player now has country\_id and current\_club\_id as foreign keys. These were implied in CP1 but not shown. Adding them makes nationality lookups and current team queries straightforward.

Game now includes competition\_id and attendance. Without competition\_id there was no way to tell which league a game belonged to, so that was a necessary fix. Attendance was added as a useful extra attribute.

PlayerValuation now includes current\_club\_id. Lets us see which club a player was at when a particular valuation was recorded, which is useful for tracking value changes around transfers.

## 2. Relational Schema

Country(country\_id, name, confederation)

- Primary key: country\_id
- Candidate keys: (name)
- Foreign keys: none

Stadium(stadium\_id, name, city, capacity)

- Primary key: stadium\_id
- Candidate keys: none
- Foreign keys: none

Competition(competition\_id, name, type)

- Primary key: competition\_id
- Candidate keys: (name)
- Foreign keys: none

Club(club\_id, name, type, squad\_size, stadium\_id, country\_id)

- Primary key: club\_id
- Candidate keys: (name)

- Foreign keys: stadium\_id references Stadium(stadium\_id), country\_id references Country(country\_id)

ClubCompetition(competed\_id, club\_id, competition\_id, placement)

- Primary key: competed\_id
- Candidate keys: (club\_id, competition\_id)
- Foreign keys: club\_id references Club(club\_id), competition\_id references Competition(competition\_id)
- Note: Bridge table resolving the M:N relationship between Club and Competition.

Player(player\_id, name, date\_of\_birth, position, sub\_position, foot, height, country\_id, current\_club\_id)

- Primary key: player\_id
- Candidate keys: none
- Foreign keys: country\_id references Country(country\_id), current\_club\_id references Club(club\_id)

Game(game\_id, date, season, round, home\_club\_id, away\_club\_id, home\_score, away\_score, competition\_id, stadium\_id, attendance)

- Primary key: game\_id
- Candidate keys: (date, home\_club\_id, away\_club\_id)
- Foreign keys: home\_club\_id references Club(club\_id), away\_club\_id references Club(club\_id), competition\_id references Competition(competition\_id), stadium\_id references Stadium(stadium\_id)

Appearance(appearance\_id, player\_id, game\_id, club\_id, minutes\_played, goals, assists, yellow\_cards, red\_cards)

- Primary key: appearance\_id
- Candidate keys: (player\_id, game\_id)
- Foreign keys: player\_id references Player(player\_id), game\_id references Game(game\_id), club\_id references Club(club\_id)
- Note: Bridge table resolving the M:N relationship between Player and Game. Contains 1.8M+ rows.

GameEvent(event\_id, game\_id, player\_id, club\_id, event\_type, minute, description)

- Primary key: event\_id
- Candidate keys: none
- Foreign keys: game\_id references Game(game\_id), player\_id references Player(player\_id), club\_id references Club(club\_id)
- Note: Weak entity, depends on Game for identification.

PlayerValuation(valuation\_id, player\_id, date, market\_value\_eur, current\_club\_id)

- Primary key: valuation\_id
- Candidate keys: (player\_id, date)
- Foreign keys: player\_id references Player(player\_id), current\_club\_id references Club(club\_id)

Transfer(transfer\_id, player\_id, from\_club\_id, to\_club\_id, transfer\_date, transfer\_fee, season)

- Primary key: transfer\_id
- Candidate keys: none
- Foreign keys: player\_id references Player(player\_id), from\_club\_id references Club(club\_id), to\_club\_id references Club(club\_id)

PlayerSeasonStats(stat\_id, player\_id, competition\_id, season, matches\_played, starts, minutes\_played, goals, assists, passing\_accuracy, progressive\_passes, shot\_creating\_actions, tackles, interceptions, blocks)

- Primary key: stat\_id
- Candidate keys: (player\_id, competition\_id, season)
- Foreign keys: player\_id references Player(player\_id), competition\_id references Competition(competition\_id)
- Note: Enrichment table from FBref secondary dataset.

### 3. Functional Dependencies

Country:

- country\_id -> name, confederation
- name -> country\_id, confederation

Stadium:

- stadium\_id -> name, city, capacity

Competition:

- competition\_id -> name, type
- name -> competition\_id, type

Club:

- club\_id -> name, type, squad\_size, stadium\_id, country\_id
- name -> club\_id, type, squad\_size, stadium\_id, country\_id

ClubCompetition:

- competed\_id -> club\_id, competition\_id, placement
- (club\_id, competition\_id) -> competed\_id, placement

Player:

- player\_id -> name, date\_of\_birth, position, sub\_position, foot, height, country\_id,  
current\_club\_id

Game:

- game\_id -> date, season, round, home\_club\_id, away\_club\_id, home\_score, away\_score,  
competition\_id, stadium\_id, attendance
- (date, home\_club\_id, away\_club\_id) -> game\_id, season, round, home\_score,  
away\_score, competition\_id, stadium\_id, attendance

Appearance:

- appearance\_id -> player\_id, game\_id, club\_id, minutes\_played, goals, assists,  
yellow\_cards, red\_cards

- (player\_id, game\_id) -> appearance\_id, club\_id, minutes\_played, goals, assists, yellow\_cards, red\_cards

GameEvent:

- event\_id -> game\_id, player\_id, club\_id, event\_type, minute, description

PlayerValuation:

- valuation\_id -> player\_id, date, market\_value\_eur, current\_club\_id
- (player\_id, date) -> valuation\_id, market\_value\_eur, current\_club\_id

Transfer:

- transfer\_id -> player\_id, from\_club\_id, to\_club\_id, transfer\_date, transfer\_fee, season

PlayerSeasonStats:

- stat\_id -> player\_id, competition\_id, season, matches\_played, starts, minutes\_played, goals, assists, passing\_accuracy, progressive\_passes, shot\_creating\_actions, tackles, interceptions, blocks
- (player\_id, competition\_id, season) -> stat\_id, matches\_played, starts, minutes\_played, goals, assists, passing\_accuracy, progressive\_passes, shot\_creating\_actions, tackles, interceptions, blocks



#### 4. Normalization Analysis

All tables are assumed to be in 1NF because each attribute contains atomic (indivisible) values, and each row is uniquely identifiable by the primary key. A table is in 2NF if it is in 1NF and every non-prime attribute is fully functionally dependent on the entire primary key, so this only becomes an issue with composite primary keys. A table is in 3NF when it is in 2NF and has no transitive dependencies, meaning a non-key attribute cannot determine another non-key attribute.

##### **Country: in 3NF**

There is a single primary key (country\_id), so 2NF is satisfied automatically. The candidate key name also determines the other attributes. There are no transitive dependencies.

##### **Stadium: in 3NF**

Single primary key (stadium\_id), so 2NF holds. name, city, and capacity all depend directly on stadium\_id. None of them determine each other (city doesn't fix capacity, capacity doesn't fix name), so there are no transitive dependencies.

##### **Competition: in 3NF**

The single primary key is competition\_id. The candidate key is name. All non-key attributes depend on the key and there are no transitive dependencies.

##### **Club: in 3NF**

Single primary key (club\_id), so 2NF is satisfied. There was a possible transitive dependency here: club\_id → stadium\_id, and stadium\_id could then determine stadium\_name, city, and capacity if those attributes lived on this table. Because we pulled stadium details into a separate

Stadium entity and only keep stadium\_id as a foreign key, that transitive dependency is removed. The candidate key name also determines all other attributes.

### **ClubCompetition: in 3NF**

This table uses a surrogate primary key (competed\_id) and also has a unique composite candidate key (club\_id, competition\_id). Because of the surrogate key, all non-key attributes are fully dependent on the entire primary key, which satisfies 2NF. The attribute placement is functionally dependent on the full candidate key (club\_id, competition\_id) rather than just part of it, so there are no partial dependencies. There is no transitive dependency since placement does not determine any other attribute.

### **Player: in 3NF**

The single primary key is player\_id, so 2NF holds. country\_id and current\_club\_id are foreign keys pointing to other tables, not transitive dependencies within this one. No non-key attribute determines another non-key attribute.

### **Game: in 3NF**

The primary key (game\_id) is a single column, so 2NF is satisfied. All non-key attributes depend on game\_id. There was a transitive dependency risk with stadium\_id, but this is resolved the same way as in Club: Game only stores stadium\_id, not stadium\_name or capacity, so the stadium details live in the Stadium table instead.

### **Appearance: in 3NF**

appearance\_id is the surrogate primary key, so all other attributes are fully dependent on it, satisfying 2NF. There is also a candidate key (player\_id, game\_id), and the stat attributes (goals, assists, minutes\_played, cards) are fully dependent on that combination. One point worth noting:

club\_id depends on the player-and-game combination rather than the player alone, because a player can represent different clubs in different games due to mid-season transfers. There are no transitive dependencies since a stat like red\_cards does not determine any other attribute.

### **GameEvent: in 3NF**

The primary key is event\_id, so 2NF holds. game\_id, player\_id, and club\_id are foreign keys pointing outward. No non-key attribute determines another, so there are no transitive dependencies.

### **PlayerValuation: in 3NF**

valuation\_id is the surrogate primary key, and (player\_id, date) is a candidate key. market\_value\_eur and current\_club\_id both depend on which player is being valued and on which date, not on each other. No transitive dependencies.

### **Transfer: in 3NF**

transfer\_id is the primary key, so 2NF is satisfied. One thing worth flagging: season could technically be derived from transfer\_date, which would make it a transitive dependency (transfer\_id → transfer\_date → season). We decided to keep season as its own stored attribute anyway. The reason is that soccer seasons don't line up cleanly with calendar dates -- a transfer in January and one the previous July both belong to the same season -- so deriving it purely from the date would need extra business logic. The tradeoff is a small amount of redundancy in exchange for simpler, unambiguous season-based queries. This is the one intentional denormalization in our design.

### **PlayerSeasonStats: in 3NF**

stat\_id is the surrogate primary key, and (player\_id, competition\_id, season) is a candidate key.

All the stat columns (passing\_accuracy, tackles, shot\_creating\_actions, etc.) depend on that full combination, since a player's numbers differ by competition and season. No stat determines another stat, so there are no transitive dependencies.

## 5. Initial Data Dictionary

The following data dictionary describes each table and attribute in the PitchStats database, including data types, meanings, key roles, data sources, and cleaning notes. The role of each table is listed alongside its name.

**Table: Country** — *Lookup / reference table*

| Attribute     | Data Type    | Meaning                                   | Key           | Source         | Notes                           |
|---------------|--------------|-------------------------------------------|---------------|----------------|---------------------------------|
| country_id    | INT          | Unique identifier for each country        | PK            | Transfer markt | —                               |
| name          | VARCHAR(100) | Country name                              | Candidate key | Transfer markt | Cleaned for consistent spelling |
| confederation | VARCHAR(20)  | FIFA confederation (UEFA, CONMEBOL, etc.) | —             | Transfer markt | —                               |

**Table: Stadium** — *Lookup / reference table*

| Attribute  | Data Type             | Meaning                               | Key | Source            | Notes                                                                       |
|------------|-----------------------|---------------------------------------|-----|-------------------|-----------------------------------------------------------------------------|
| stadium_id | INT<br>AUTO_INCREMENT | Unique identifier<br>for each stadium | PK  | Derived           | Generated<br>by<br>deduplicating stadium<br>names from<br>game/club<br>data |
| name       | VARCHAR(150)          | Stadium name                          | —   | Transfer<br>markt | —                                                                           |
| city       | VARCHAR(100)          | City the stadium<br>is located in     | —   | Transfer<br>markt | NULL<br>where not<br>available                                              |
| capacity   | INT                   | Seating capacity                      | —   | Transfer<br>markt | NULL if not<br>recorded                                                     |

**Table: Competition — Core entity table**

| Attribute      | Data Type         | Meaning                               | Key           | Source        | Notes                         |
|----------------|-------------------|---------------------------------------|---------------|---------------|-------------------------------|
| competition_id | VARCHAR<br>R(10)  | Unique identifier for the competition | PK            | Transfermarkt | String codes like "GB1", "CL" |
| name           | VARCHAR<br>R(100) | Full competition name                 | Candidate key | Transfermarkt | —                             |
| type           | VARCHAR<br>R(20)  | Category (league, cup, international) | —             | Transfermarkt | Categorized from source       |

**Table: Club — Core entity table**

| Attribute | Data Type         | Meaning                         | Key           | Source        | Notes                 |
|-----------|-------------------|---------------------------------|---------------|---------------|-----------------------|
| club_id   | INT               | Unique identifier for each club | PK            | Transfermarkt | —                     |
| name      | VARCHAR<br>R(100) | Club name                       | Candidate key | Transfermarkt | —                     |
| type      | VARCHAR<br>R(30)  | Club category                   | —             | Transfermarkt | e.g., domestic, youth |

|            |     |                                |              |                |                 |
|------------|-----|--------------------------------|--------------|----------------|-----------------|
| squad_size | INT | Number of players in the squad | —            | Transfer markt | Latest snapshot |
| stadium_id | INT | Club's home stadium            | FK → Stadium | Transfer markt | —               |
| country_id | INT | Country the club is based in   | FK → Country | Transfer markt | —               |

**Table: ClubCompetition — Bridge table**

| Attribute      | Data Type             | Meaning                          | Key              | Source         | Notes              |
|----------------|-----------------------|----------------------------------|------------------|----------------|--------------------|
| competed_id    | INT<br>AUTO_INCREMENT | Unique identifier for each entry | PK               | Generated      | Surrogate key      |
| club_id        | INT                   | Club that competed               | FK → Club        | Transfer markt | —                  |
| competition_id | VARCHAR(10)           | Competition they took part in    | FK → Competition | Transfer markt | —                  |
| placement      | VARCHAR(20)           | Final standing or result         | —                | Transfer markt | e.g., "1", "Group" |

|  |  |  |  |  |                               |
|--|--|--|--|--|-------------------------------|
|  |  |  |  |  | Stage",<br>NULL if<br>ongoing |
|--|--|--|--|--|-------------------------------|

**Table: Player** — *Core entity table*

| Attribute     | Data Type         | Meaning                              | Key | Source            | Notes                                           |
|---------------|-------------------|--------------------------------------|-----|-------------------|-------------------------------------------------|
| player_id     | INT               | Unique identifier<br>for each player | PK  | Transfer<br>markt | —                                               |
| name          | VARCHAR<br>R(150) | Player's full name                   | —   | Transfer<br>markt | —                                               |
| date_of_birth | DATE              | Birth date                           | —   | Transfer<br>markt | Used in<br>age-based<br>queries                 |
| position      | VARCHAR<br>R(30)  | General position                     | —   | Transfer<br>markt | Attack,<br>Midfield,<br>Defender,<br>Goalkeeper |
| sub_position  | VARCHAR<br>R(50)  | Specific role                        | —   | Transfer<br>markt | e.g., Right<br>Winger,<br>Centre-Back           |



|                 |             |                       |              |               |                                       |
|-----------------|-------------|-----------------------|--------------|---------------|---------------------------------------|
| foot            | VARCHAR(10) | Preferred foot        | —            | Transfermarkt | Left, Right, Both; NULL if unknown    |
| height          | INT         | Height in centimeters | —            | Transfermarkt | Converted from string during cleaning |
| country_id      | INT         | Player's nationality  | FK → Country | Transfermarkt | —                                     |
| current_club_id | INT         | Player's current club | FK → Club    | Transfermarkt | Updates on transfer                   |

**Table: Game** — *Core entity table*

| Attribute | Data Type | Meaning                          | Key | Source        | Notes                                |
|-----------|-----------|----------------------------------|-----|---------------|--------------------------------------|
| game_id   | INT       | Unique identifier for each match | PK  | Transfermarkt | —                                    |
| date      | DATE      | Date the match was played        | —   | Transfermarkt | Key attribute for time-based queries |

|                    |                  |                                       |                         |                   |                                        |
|--------------------|------------------|---------------------------------------|-------------------------|-------------------|----------------------------------------|
| season             | INT              | Season start year<br>(2023 = 2023/24) | —                       | Transfer<br>markt | —                                      |
| round              | VARCHAR<br>R(50) | Matchday or<br>round label            | —                       | Transfer<br>markt | e.g., "1",<br>"Quarter-Finals"         |
| home_club<br>_id   | INT              | Club playing at<br>home               | FK →<br>Club            | Transfer<br>markt | Determines<br>home/away<br>designation |
| away_club<br>_id   | INT              | Club playing<br>away                  | FK →<br>Club            | Transfer<br>markt | Determines<br>home/away<br>designation |
| home_scor<br>e     | INT              | Goals scored by<br>home team          | —                       | Transfer<br>markt | —                                      |
| away_scor<br>e     | INT              | Goals scored by<br>away team          | —                       | Transfer<br>markt | —                                      |
| competitio<br>n_id | VARCHAR<br>R(10) | Competition the<br>game belongs to    | FK →<br>Competi<br>tion | Transfer<br>markt | —                                      |

|            |     |                                 |              |                 |                               |
|------------|-----|---------------------------------|--------------|-----------------|-------------------------------|
| stadium_id | INT | Venue where the game was played | FK → Stadium | Transfer market | Consolidated per CP1 feedback |
| attendance | INT | Number of spectators            | —            | Transfer market | NULL if not recorded          |

**Table: Appearance** — *Bridge / fact table (1.8M+ rows)*

| Attribute      | Data Type             | Meaning                               | Key         | Source          | Notes                            |
|----------------|-----------------------|---------------------------------------|-------------|-----------------|----------------------------------|
| appearance_id  | INT<br>AUTO_INCREMENT | Unique identifier for each appearance | PK          | Generated       | Surrogate key                    |
| player_id      | INT                   | Player who appeared                   | FK → Player | Transfer market | —                                |
| game_id        | INT                   | Game the player appeared in           | FK → Game   | Transfer market | —                                |
| club_id        | INT                   | Club the player represented           | FK → Club   | Transfer market | Matters for mid-season transfers |
| minutes_played | INT                   | Minutes on the pitch                  | —           | Transfer market | 0 if unused substitute           |

|              |     |                          |   |                   |        |
|--------------|-----|--------------------------|---|-------------------|--------|
| goals        | INT | Goals scored             | — | Transfer<br>markt | —      |
| assists      | INT | Assists recorded         | — | Transfer<br>markt | —      |
| yellow_cards | INT | Yellow cards<br>received | — | Transfer<br>markt | 0 or 1 |
| red_cards    | INT | Red cards<br>received    | — | Transfer<br>markt | 0 or 1 |

**Table: GameEvent** — *Fact table (weak entity)*

| Attribute | Data Type             | Meaning                             | Key            | Source            | Notes                               |
|-----------|-----------------------|-------------------------------------|----------------|-------------------|-------------------------------------|
| event_id  | INT<br>AUTO_INCREMENT | Unique identifier<br>for each event | PK             | Generated         | Surrogate<br>key for weak<br>entity |
| game_id   | INT                   | Game the event<br>occurred in       | FK →<br>Game   | Transfer<br>markt | Identifying<br>relationship         |
| player_id | INT                   | Player involved                     | FK →<br>Player | Transfer<br>markt | NULL for<br>non-player<br>events    |

|             |              |                           |           |                 |                             |
|-------------|--------------|---------------------------|-----------|-----------------|-----------------------------|
| club_id     | INT          | Club the player was on    | FK → Club | Transfer market | —                           |
| event_type  | VARCHAR(30)  | Type of event             | —         | Transfer market | Goals, Cards, Substitutions |
| minute      | INT          | Minute the event occurred | —         | Transfer market | Extra time stored as 90+    |
| description | VARCHAR(255) | Extra detail              | —         | Transfer market | e.g., penalty, own goal     |

**Table: PlayerValuation** — *Fact / transaction table*

| Attribute    | Data Type             | Meaning             | Key         | Source          | Notes |
|--------------|-----------------------|---------------------|-------------|-----------------|-------|
| valuation_id | INT<br>AUTO_INCREMENT | Unique identifier   | PK          | Generated       | —     |
| player_id    | INT                   | Player being valued | FK → Player | Transfer market | —     |

|                  |        |                                |           |                |                                          |
|------------------|--------|--------------------------------|-----------|----------------|------------------------------------------|
| date             | DATE   | Date of the valuation snapshot | —         | Transfer markt | Enables market value time-series queries |
| market_value_eur | BIGINT | Market value in euros          | —         | Transfer markt | Stored as integer; NULL if unknown       |
| current_club_id  | INT    | Club at time of valuation      | FK → Club | Transfer markt | —                                        |

**Table: Transfer** — *Fact / transaction table*

| Attribute    | Data Type             | Meaning                | Key         | Source         | Notes                |
|--------------|-----------------------|------------------------|-------------|----------------|----------------------|
| transfer_id  | INT<br>AUTO_INCREMENT | Unique identifier      | PK          | Generated      | —                    |
| player_id    | INT                   | Player who transferred | FK → Player | Transfer markt | —                    |
| from_club_id | INT                   | Club the player left   | FK → Club   | Transfer markt | NULL for free agents |

|               |             |                        |           |                |                                           |
|---------------|-------------|------------------------|-----------|----------------|-------------------------------------------|
| to_club_id    | INT         | Club the player joined | FK → Club | Transfer markt | NULL if retiring                          |
| transfer_date | DATE        | Date of the transfer   | —         | Transfer markt | —                                         |
| transfer_fee  | BIGINT      | Fee in euros           | —         | Transfer markt | 0 for free transfers, NULL if undisclosed |
| season        | VARCHAR(10) | Season of the transfer | —         | Transfer markt | Stored intentionally (see normalization)  |

**Table: PlayerSeasonStats** — *Enrichment table (secondary dataset)*

| Attribute | Data Type             | Meaning           | Key | Source    | Notes |
|-----------|-----------------------|-------------------|-----|-----------|-------|
| stat_id   | INT<br>AUTO_INCREMENT | Unique identifier | PK  | Generated | —     |

|                  |              |                              |                  |       |                                          |
|------------------|--------------|------------------------------|------------------|-------|------------------------------------------|
| player_id        | INT          | Player these stats belong to | FK → Player      | FBref | Matched to Transfermarkt via name + club |
| competition_id   | VARCHAR(10)  | Competition for these stats  | FK → Competition | FBref | Mapped to our competition_id values      |
| season           | INT          | Season year                  | —                | FBref | —                                        |
| matches_played   | INT          | Matches played that season   | —                | FBref | —                                        |
| starts           | INT          | Games started                | —                | FBref | —                                        |
| minutes_played   | INT          | Total minutes                | —                | FBref | —                                        |
| goals            | INT          | Goals scored                 | —                | FBref | —                                        |
| assists          | INT          | Assists                      | —                | FBref | —                                        |
| passing_accuracy | DECIMAL(5,2) | Pass completion percentage   | —                | FBref | e.g., 87.50                              |



|                           |     |                                                    |   |       |                    |
|---------------------------|-----|----------------------------------------------------|---|-------|--------------------|
| progressive_<br>passes    | INT | Passes moving the<br>ball forward<br>significantly | — | FBref | Advanced<br>metric |
| shot_creatin<br>g_actions | INT | Actions leading to<br>a shot                       | — | FBref | Advanced<br>metric |
| tackles                   | INT | Tackles made                                       | — | FBref | Defensive<br>stat  |
| interceptions             | INT | Interceptions                                      | — | FBref | Defensive<br>stat  |
| blocks                    | INT | Shots/passes<br>blocked                            | — | FBref | Defensive<br>stat  |

## 6. Initial SQL Table Creation Plan

```
CREATE DATABASE IF NOT EXISTS pitchstats;
```

```
USE pitchstats;
```

```
CREATE TABLE Country (  
    country_id INT PRIMARY KEY,  
    name VARCHAR(100) NOT NULL UNIQUE,  
    confederation VARCHAR(20)  
);
```

```
CREATE TABLE Stadium (  
    stadium_id INT AUTO_INCREMENT PRIMARY KEY,  
    name VARCHAR(150) NOT NULL,  
    city VARCHAR(100),  
    capacity INT  
);
```

```
CREATE TABLE Competition (  
    competition_id VARCHAR(10) PRIMARY KEY,  
    name VARCHAR(100) NOT NULL UNIQUE,  
    type VARCHAR(20) NOT NULL  
);
```

```
CREATE TABLE Club (  
    club_id INT PRIMARY KEY,  
    name VARCHAR(100) NOT NULL,  
    type VARCHAR(30),  
    squad_size INT,  
    stadium_id INT,  
    country_id INT,  
    FOREIGN KEY (stadium_id) REFERENCES Stadium(stadium_id),  
    FOREIGN KEY (country_id) REFERENCES Country(country_id)  
);
```

```
CREATE TABLE ClubCompetition (  
    competed_id INT AUTO_INCREMENT PRIMARY KEY,  
    club_id INT NOT NULL,  
    competition_id VARCHAR(10) NOT NULL,  
    placement VARCHAR(20),  
    FOREIGN KEY (club_id) REFERENCES Club(club_id),  
    FOREIGN KEY (competition_id) REFERENCES Competition(competition_id),  
    UNIQUE (club_id, competition_id)  
);
```

```
CREATE TABLE Player (  

```

```

player_id INT PRIMARY KEY,
name VARCHAR(150) NOT NULL,
date_of_birth DATE,
position VARCHAR(30),
sub_position VARCHAR(50),
foot VARCHAR(10),
height INT,
country_id INT,
current_club_id INT,
FOREIGN KEY (country_id) REFERENCES Country(country_id),
FOREIGN KEY (current_club_id) REFERENCES Club(club_id)
);

```

```

CREATE TABLE Game (
    game_id INT PRIMARY KEY,
    date DATE NOT NULL,
    season INT NOT NULL,
    round VARCHAR(50),
    home_club_id INT NOT NULL,
    away_club_id INT NOT NULL,
    home_score INT,
    away_score INT,
    competition_id VARCHAR(10) NOT NULL,

```

```

    stadium_id INT,
    attendance INT,
    FOREIGN KEY (home_club_id) REFERENCES Club(club_id),
    FOREIGN KEY (away_club_id) REFERENCES Club(club_id),
    FOREIGN KEY (competition_id) REFERENCES Competition(competition_id),
    FOREIGN KEY (stadium_id) REFERENCES Stadium(stadium_id),
    CHECK (home_club_id != away_club_id)
);

```

```

CREATE TABLE Appearance (
    appearance_id INT AUTO_INCREMENT PRIMARY KEY,
    player_id INT NOT NULL,
    game_id INT NOT NULL,
    club_id INT NOT NULL,
    minutes_played INT DEFAULT 0,
    goals INT DEFAULT 0,
    assists INT DEFAULT 0,
    yellow_cards INT DEFAULT 0,
    red_cards INT DEFAULT 0,
    FOREIGN KEY (player_id) REFERENCES Player(player_id),
    FOREIGN KEY (game_id) REFERENCES Game(game_id),
    FOREIGN KEY (club_id) REFERENCES Club(club_id),
    UNIQUE (player_id, game_id)
);

```

);

```
CREATE TABLE GameEvent (  
    event_id INT AUTO_INCREMENT PRIMARY KEY,  
    game_id INT NOT NULL,  
    player_id INT,  
    club_id INT,  
    event_type VARCHAR(30) NOT NULL,  
    minute INT,  
    description VARCHAR(255),  
    FOREIGN KEY (game_id) REFERENCES Game(game_id),  
    FOREIGN KEY (player_id) REFERENCES Player(player_id),  
    FOREIGN KEY (club_id) REFERENCES Club(club_id)  
);
```

```
CREATE TABLE PlayerValuation (  
    valuation_id INT AUTO_INCREMENT PRIMARY KEY,  
    player_id INT NOT NULL,  
    date DATE NOT NULL,  
    market_value_eur BIGINT,  
    current_club_id INT,  
    FOREIGN KEY (player_id) REFERENCES Player(player_id),  
    FOREIGN KEY (current_club_id) REFERENCES Club(club_id),
```

```
    UNIQUE (player_id, date)
);
```

```
CREATE TABLE Transfer (
    transfer_id INT AUTO_INCREMENT PRIMARY KEY,
    player_id INT NOT NULL,
    from_club_id INT,
    to_club_id INT,
    transfer_date DATE,
    transfer_fee BIGINT DEFAULT 0,
    season VARCHAR(10),
    FOREIGN KEY (player_id) REFERENCES Player(player_id),
    FOREIGN KEY (from_club_id) REFERENCES Club(club_id),
    FOREIGN KEY (to_club_id) REFERENCES Club(club_id)
);
```

```
CREATE TABLE PlayerSeasonStats (
    stat_id INT AUTO_INCREMENT PRIMARY KEY,
    player_id INT NOT NULL,
    competition_id VARCHAR(10) NOT NULL,
    season INT NOT NULL,
    matches_played INT DEFAULT 0,
    starts INT DEFAULT 0,
```

```

minutes_played INT DEFAULT 0,

goals INT DEFAULT 0,

assists INT DEFAULT 0,

passing_accuracy DECIMAL(5,2),

progressive_passes INT DEFAULT 0,

shot_creating_actions INT DEFAULT 0,

tackles INT DEFAULT 0,

interceptions INT DEFAULT 0,

blocks INT DEFAULT 0,

FOREIGN KEY (player_id) REFERENCES Player(player_id),

FOREIGN KEY (competition_id) REFERENCES Competition(competition_id),

UNIQUE (player_id, competition_id, season)

);

```

## 7. Testing

```

mysql> USE pitchstats;
Database changed
mysql> SHOW TABLES;
+-----+
| Tables_in_pitchstats |
+-----+
| Appearance           |
| Club                 |
| ClubCompetition      |
| Competition          |
| Country              |
| Game                 |
| GameEvent            |
| Player               |
| PlayerSeasonStats    |
| PlayerValuation      |
| Stadium              |
| Transfer             |
+-----+
12 rows in set (0.00 sec)

```



```
mysql> INSERT INTO Country VALUES (1, 'England', 'UEFA');
Query OK, 1 row affected (0.01 sec)
mysql> INSERT INTO Country VALUES (2, 'Spain', 'UEFA');
Query OK, 1 row affected (0.00 sec)
mysql> INSERT INTO Country VALUES (3, 'Brazil', 'CONMEBOL');
Query OK, 1 row affected (0.00 sec)
mysql> INSERT INTO Stadium (name, city, capacity)
-> VALUES ('Old Trafford', 'Manchester', 74310);
Query OK, 1 row affected (0.01 sec)
mysql> INSERT INTO Competition VALUES ('GB1', 'Premier League', 'league');
Query OK, 1 row affected (0.00 sec)
mysql> INSERT INTO Club VALUES (1, 'Manchester United', 'domestic', 25, 1, 1);
Query OK, 1 row affected (0.01 sec)
mysql> INSERT INTO Player VALUES (1, 'Bruno Fernandes', '1994-09-08',
-> 'Midfield', 'Attacking Midfield', 'Right', 179, 1, 1);
Query OK, 1 row affected (0.00 sec)

mysql> SELECT * FROM Player;
+-----+-----+-----+-----+
| player_id | name           | date_of_birth | position |
+-----+-----+-----+-----+
|          1 | Bruno Fernandes | 1994-09-08    | Midfield |
+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql>
```

```
-- TEST: PRIMARY KEY constraint (duplicate id)
mysql> INSERT INTO Country VALUES (1, 'France', 'UEFA');
ERROR 1062 (23000): Duplicate entry '1' for key 'Country.PRIMARY'

-- TEST: UNIQUE constraint (duplicate name)
mysql> INSERT INTO Country VALUES (4, 'England', 'UEFA');
ERROR 1062 (23000): Duplicate entry 'England' for key 'Country.name'

-- TEST: FOREIGN KEY constraint (country_id 999 does not exist)
mysql> INSERT INTO Club VALUES (99, 'Fake Club', 'domestic', 20, 1, 999);
ERROR 1452 (23000): Cannot add or update a child row: a foreign key
constraint fails (`pitchstats`.`Club`, CONSTRAINT `Club_ibfk_2` FOREIGN
KEY (`country_id`) REFERENCES `Country` (`country_id`))

-- TEST: CHECK constraint (home team == away team)
mysql> INSERT INTO Game VALUES (1, '2024-01-15', 2023, '20', 1, 1, 2,
-> 1, 'GB1', 1, 74000);
ERROR 3819 (HY000): Check constraint 'Game_chk_1' is violated.

mysql>
```

## 8. Updates to Technical Plan or Project Management

*No changes to the technical plan or project management plan at this time.*

## 9. AI-Use Statement

- Our team used Claude (Anthropic) as an AI tool for this checkpoint.
- We used it to revise the ER diagram and to ideate how to address the changes suggested in the Checkpoint 1 feedback (consolidating the stadium information, clarifying the GameEvent connections, and incorporating the secondary dataset).

- We reviewed each suggested fix ourselves and decided which ones to apply, adjusting the design to make sure it matched our actual datasets and met the project requirements.
- All changes were verified by team members before finalizing the diagram and schema.